Worksheet 9.4

Metamorphic rocks

Answers

Rocks — the new generation

You will need:
labelled samples of granite, gneiss, limestone, marble, sandstone, quartzite, shale and slate
hand lens

- Try to sort the rocks into pairs of 'parent' rock and corresponding metamorphic rock.

1. Why is the term 'parent' rock used? Answers may vary. Just as a parent produces a child, a parent rock is the basis of the new metamorphic rock.

- Examine each pair of rocks with a hand lens.
- Complete the table below by noting the similarities and differences between the 'parent' and metamorphic rock of each pair.

2. Use the last column of your table to suggest whether the main cause of metamorphism was heat or pressure.

Comparing 'parent' and metamorphic rocks

'Parent' rock	Metamorphic rock	Similarities	Differences	Main cause of metamorphism
Shale	Slate	Both have similar colours and visible layers.	Slate is harder than shale and fractures into flat sheets.	Pressure
Granite	Gneiss	Both are made up of crystals of mica (black), quartz (white or clear) and feldspar (grey, pink or beige).	Gneiss has smaller crystals and has bands.	Pressure
Sandstone	Quartzite	Both have silica grains evident.	Quartzite is harder and has a more sugary, crystalline appearance.	Heat
Limestone	Marble	Both are pale and made of calcium carbonate.	Limestone is more porous and lighter than marble; marble is harder and more crystalline.	Heat

© John Wiley & Sons Australia, Ltd